

MACHINE LEARNING PROJECT

Match Outcome Prediction Using Logistic Regression and Random Forest

M. Sai Ajith Reddy · 2520090186

G. V. Karthik Reddy · 2520080006

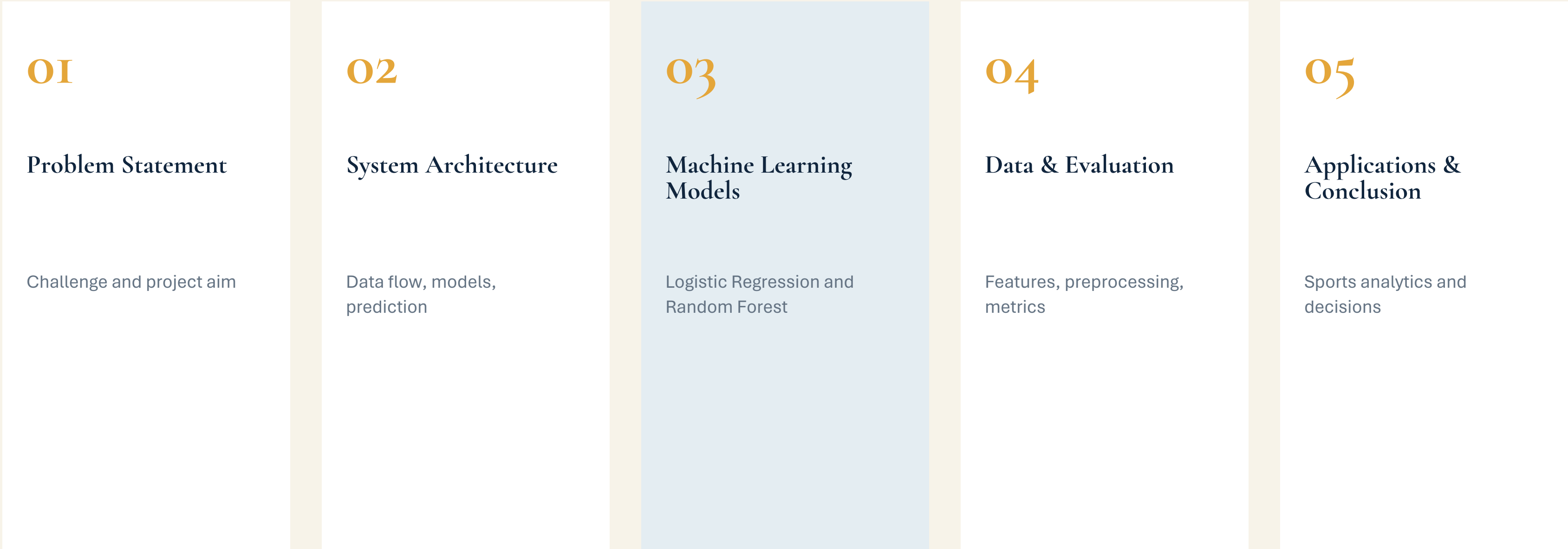
GUIDE Dr. Lakshmi Presanna
Assistant Professor, AI&ML

Department of Artificial Intelligence & Machine Learning



ML Structure of This Presentation

A clear path from problem framing to practical application



I • Problem Statement

THE CHALLENGE

- Match outcomes depend on many performance factors
- Team statistics and previous records shape prediction
- Player performance and venue conditions matter
- Win-loss history must be analysed
- Manual analysis is difficult

OUR OBJECTIVE

Build a machine learning system that analyses historical and performance data to predict a binary win / lose outcome.



MULTIPLE

Performance Factors

HISTORICAL

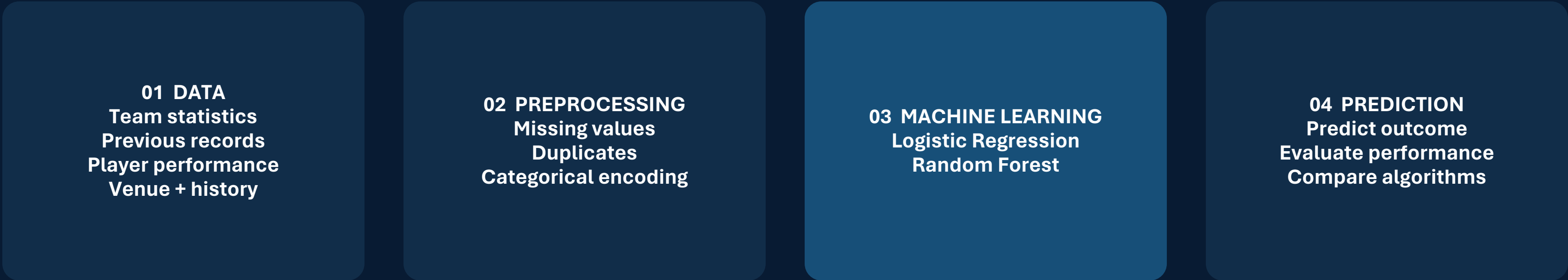
Match Data

BINARY

Win / Lose Prediction

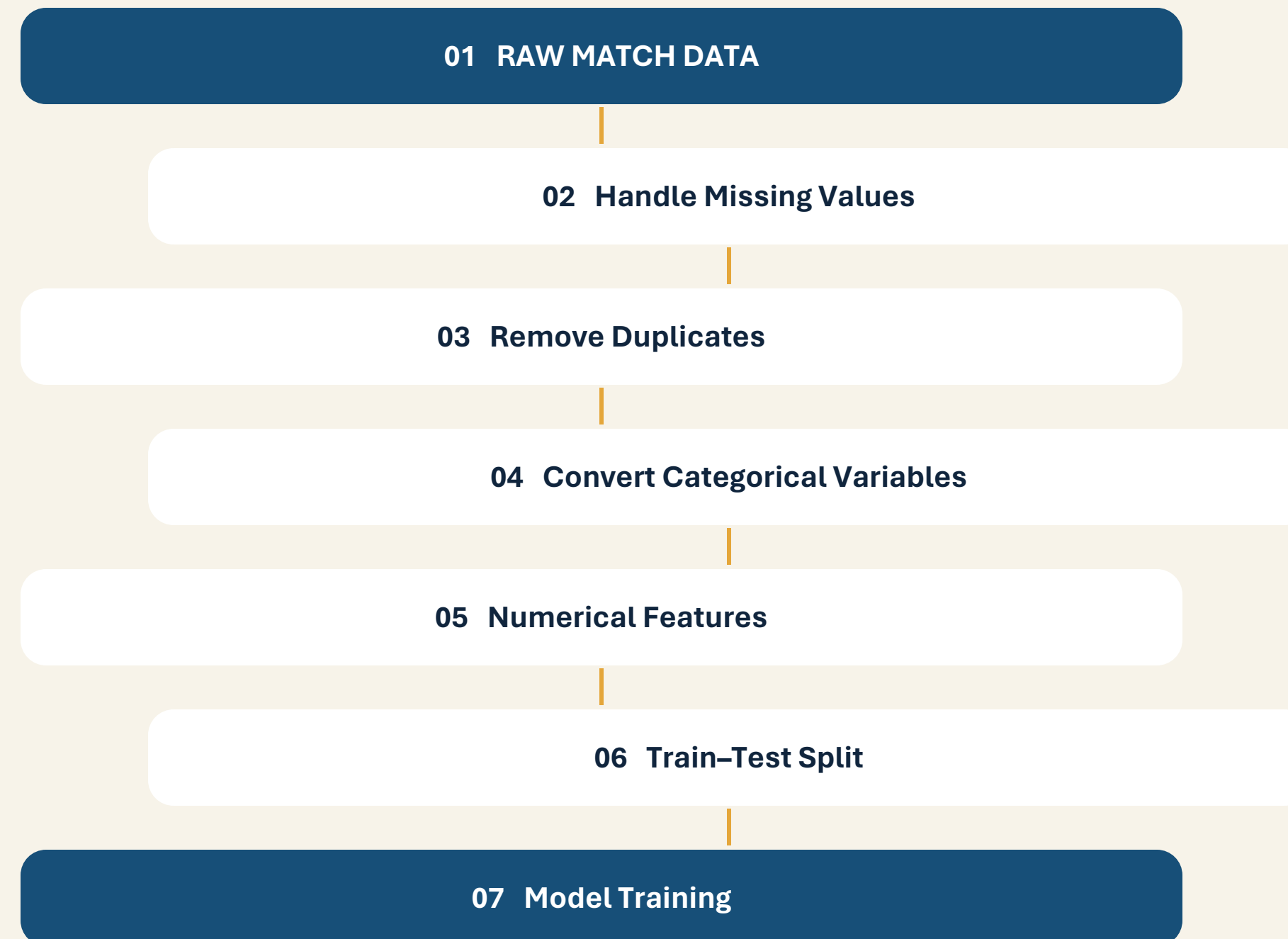
2 · System Architecture

From historical records to evaluated predictions



3 · Data Preprocessing

A disciplined pipeline converts raw records into model-ready features.



Why Preprocessing?

- ✓ Improves data quality
- ✓ Makes data suitable for ML
- ✓ Removes duplicate or incomplete information
- ✓ Converts categories to numerical values

3 · Machine Learning Models

LOGISTIC REGRESSION

- Supervised learning algorithm
- Baseline model
- Suitable for binary classification
- Classifies win / loss outcomes
- Simple and effective

Logistic Regression → Classification

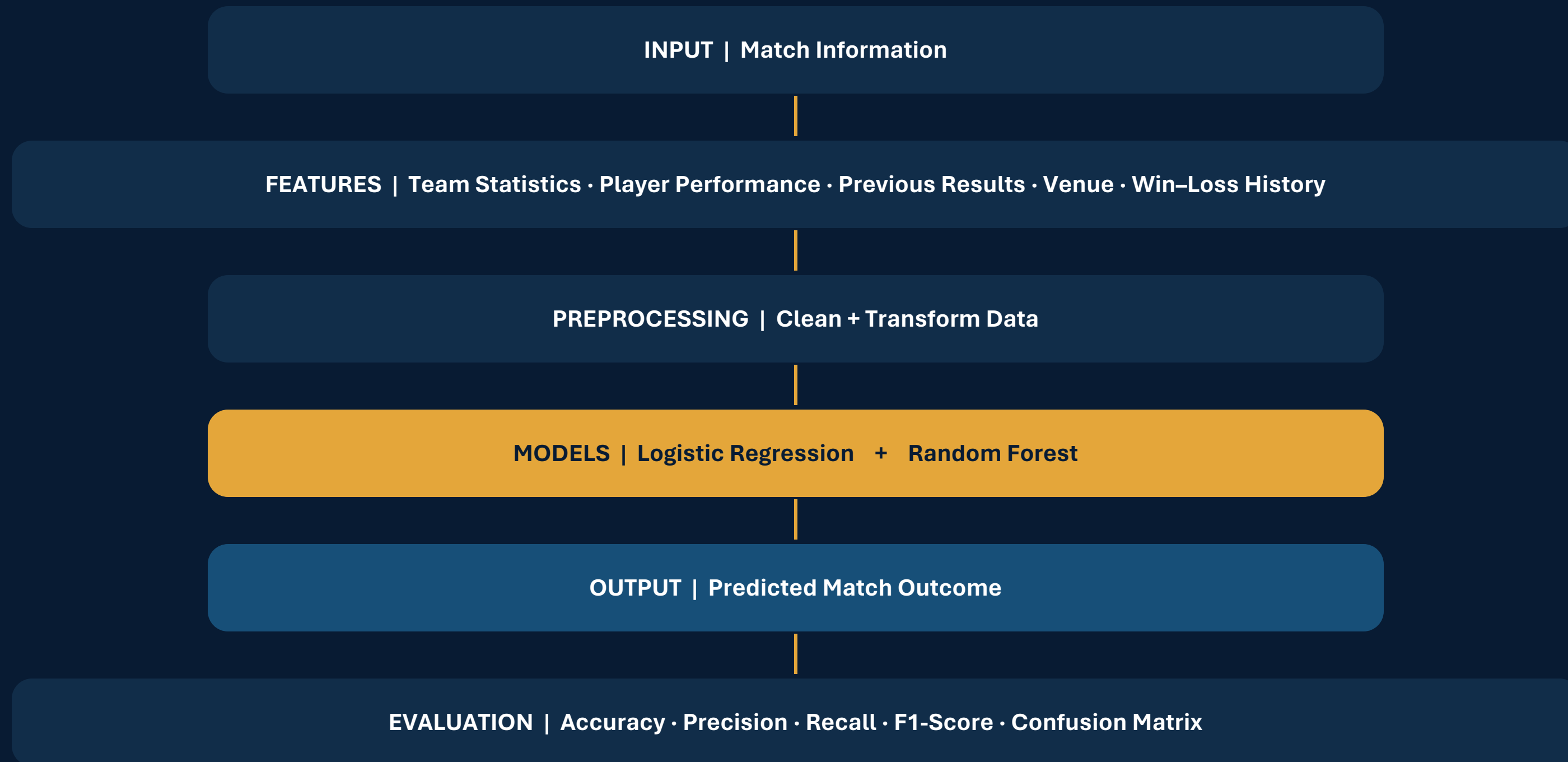
RANDOM FOREST

- Ensemble learning algorithm
- Combines multiple decision trees
- Captures complex feature relationships
- Improves predictive performance
- Helps reduce overfitting

Multiple Decision Trees → Prediction

4 · Prediction Workflow

One continuous path from match context to measurable performance



4 • Model Evaluation

Five complementary views of classification quality — without cherry-picking a single metric.

ACCURACY

Overall proportion of correct predictions

PRECISION

How many predicted positives are actually positive

RECALL

How effectively positive outcomes are identified

F1-SCORE

Combines precision and recall

CONFUSION MATRIX

Shows correct and incorrect classifications

Predicted Win	Predict	
Actual Win		
Actual Loss		

	TRUE POSITIVE	FALSE NEGATIVE
	FALSE POSITIVE	TRUE NEGATIVE

5 · Logistic Regression vs Random Forest

Comparison purpose: evaluate both models with the same metrics — no winner is declared in advance.

CATEGORY	LOGISTIC REGRESSION	RANDOM FOREST
Model type	Supervised classification	Ensemble learning
Role	Baseline model	Comparative / improved model
Structure	Logistic classification	Multiple decision trees
Relationship handling	Suitable for simpler relationships	Captures complex relationships
Evaluation	Accuracy, Precision, Recall, F1	Accuracy, Precision, Recall, F1

5 · Applications of the System

Where probabilistic match insight creates practical value

01 SPORTS ANALYTICS

Analyse historical sports data

02 TEAM ANALYSIS

Understand team performance

03 STRATEGY PLANNING

Support data-driven planning

04 MATCH ANALYSIS

Study performance factors

05 ANALYST SUPPORT

Assist evaluation of teams and matches

06 FAN INSIGHTS

Understand match probabilities

Project Contribution

BENEFITS

- Data-driven match prediction
- Analysis of multiple performance factors
- Comparison of two ML algorithms
- Practical supervised learning application
- Experience with preprocessing and evaluation

TECHNOLOGIES

Python

Scikit-learn

Pandas + NumPy

Matplotlib + Seaborn

CSV datasets

Jupyter / Colab / VS Code

GitHub

Conclusion and Future Scope

WHAT WE DEVELOPED

- ML approach for sports match outcome prediction
- Team stats, records, player performance, venue and history
- Logistic Regression baseline model
- Random Forest ensemble model
- Accuracy, precision, recall, F1 and confusion matrix

FUTURE SCOPE

- Use larger, more diverse sports datasets
- Include additional performance features
- Experiment with more ML algorithms
- Improve reliability with deeper history
- Develop a user-friendly prediction interface

THANK YOU

Questions?